

Fnu Mohbat

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SUMMARY

Ph.D. in Computer Science with strong research and applied experience in small and large language models (SLMs & LLMs), multi-modal learning, knowledge graphs (KGs), and graph neural networks (GNNs). Experienced in designing, training, and deploying LLMs, VLMs, and RAG systems on multi-GPU clusters.

Seeking a applied researcher role to contribute to cutting-edge AI development in NLP, CV, or multi-modal systems.

SKILLS SUMMARY

Generative AI: LLMs, SLMs, RAG, GraphRAG, Agentic AI, MCP, LangChain, LlamaIndex, Prompt Engineering

Machine Learning: PyTorch, TensorFlow, Keras, LoRA, RLHF (DPO, PPO, GRPO), Knowledge Graphs

Programming & Data: Python, C/C++, Pandas, NumPy, SciPy, FAISS, ChromaDB, Git, Docker, MLflow

Cloud & Systems: AWS, Azure, Palantir Foundry, HPC, OpenCV, ROS, IBM Power9

EDUCATION

Ph.D. Computer Science, Rensselaer Polytechnic Institute, Troy, NY 01/2021 - 08/2025

Dissertation: Knowledge Graph Enhanced Large Language Models for Food Understanding.

Advisor: Mohammed J. Zaki

MS Electrical Engineering, Lahore University of Management Sciences 08/2016 - 06/2018

Thesis: Content Based Image Retrieval Through Deep Learning.

BS Electrical Engineering, COMSATS Institute of Information Technology 09/2009 - 06/2014

Undergraduate Exchange, Ivy Tech Community College, Fort Wayne, IN 08/2012 - 05/2013

EXPERIENCE

TWG Global AI Santa Monica, CA

Sr. Associate, AI Researcher & Developer,

08/2025 - Present

- Fine-tuned and evaluated a suite of SLMs against frontier LLM baselines, deploying to production on Palantir Foundry—cutting inference cost 15–20x while enabling on-premise/private deployment for sensitive data.
- **SLM Tasks:** financial QA, metric extraction, loan tape parsing (“cracking”), formula generation—improving accuracy from ~95% to 99.99% and reducing manual review from ~15% to under 0.5%.
- Built a video intelligence pipeline for sports event detection, video asset management, and knowledge-graph-grounded video question answering (VQA).
- Designed a news analysis system for summarization, aggregation, and personalized tracking, enabling real-time, investor-specific insight generation at scale.
- Designed a tunable RAG framework across multiple frontier models (Claude, GPT, Gemini) that adapts retrieval and prompting strategy per use-case based on required accuracy, cost constraints, and failure severity, resulting in an optimized RAG pipeline for downstream use-cases.

Rensselaer Polytechnic Institute

Troy, NY

Graduate Research Assistant, Advisor: Mohammed J. Zaki

08/2021 - 08/2025

- Designed a knowledge-graph-augmented Retrieval-Augmented Generation (RAG) framework to improve grounding and factual accuracy of Large Language Models (LLMs), achieving a 30 – 50% improvement in recommendation relevance over baseline retrieval methods.
- Proposed multistage instruction tuning and custom loss for improving language vision models, resulting a 16-point gain in SacreBLEU and a 6-point increase in ROUGE-L scores.
- Improved document understanding, caption generation, and text summarization through LLM (LLaMA, Phi, Mistral) and Multi-Modal Models (LLaVA, miniGPT).
- Facilitated collaboration between IBM Research and Rensselaer Polytechnic Institute, combining academic and industry expertise to foster innovation (2021-2023).

Nokia Bell Labs

New Providence, NJ

ML Research Intern,

05/2025 - 08/2025

- Developed agentic AI systems to improve customer experience and assist in fault analysis and resolution, leveraging domain-specific data such as time series, customer feedback, and system logs.

IBM Thomas J. Watson Research Center

Remote

Summer Extern, Mentors: Keerthiram Murugesan

05/2023 - 08/2023

- Investigated the impact of Stable Diffusion (SD) generated images on visual concept manipulation in Multi-Modal Models (MMMs), revealing that biased visuals exacerbate toxicity and bias in generated text compared to unbiased visual and text-only inputs.

IBM Thomas J. Watson Research Center

Yorktown Heights, NY

Summer Extern, Manager: Ashish Verma

05/2022 - 08/2022

- Improved generalization of document AI models by 15 – 30% on out-of-distribution (OOD) data through novel Transformer-GNN integration, reducing OOD detection error rates and enhancing real-world adaptability.

IBM Thomas J. Watson Research Center

Remote

Summer Extern, Mentors: Ashish Verma, Catherine Finegan-Dollak

05/2021 - 08/2021

- Fine tuned transformer models and object detection models for document text understanding tasks including classification and key-value prediction.
- Annotated a proprietary visual document dataset and fine-tuned and evaluated models (Bert, LayoutLM) on it.

Lahore University of Management Sciences

Lahore, Pakistan

Research Associate in National Agriculture Robotics Lab

04/2019 - 12/2020

- Led 5-10 researchers in developing deep learning and IoT solutions for water quality assessment via foreign object detection, implemented on Raspberry Pi and NVIDIA Jetson Nano.
- Researched model compression techniques, developing a Teacher-Class network that significantly enhanced accuracy and reduced parameters by 10-30x compared to traditional single-student methods.
- Co-supervised 4 undergraduate and 2 master's students, drafted proposal for research grant proposals and industrial collaborations.

Lahore University of Management Sciences

Lahore, Pakistan

Research Assistant in Computer Vision & Graphics Lab

06/2017 - 03/2019

- Proposed a residual dyad network that amplified image retrieval by 22 mAP points on remote sensing images.
- Developed fixed-wing V-TOL UAV for autonomous target identification, payload delivery (5kg, 10km range), with 60-min flight time.
- Wrote modification codes for GTA-V to collect 100,000 road scene images with car parameters, and trained Convolutional Neural Networks (CNNs) to predict speed and steering angles for autonomous vehicle control within the game environment.
- Conducted tutorials in Computer Vision and Deep Learning classes and mentored senior year projects.

Agatos Pvt. Pakistan, Automation Engineer

08/2014 - 10/2016

- Analyzed project requirements, programmed PLCs, HMI/SCADA systems, and commissioned on site, and participated in sales meetings with customers.

PUBLICATIONS

11. KERL: Knowledge-Enhanced Personalized Recipe Recommendation using Large Language Models, Fnu Mohbat & Mohammed J. Zaki. The Association for Computational Linguistics (ACL), 2025.
10. Knowledge Graph-Enhanced LLM for Food Recommendation through Question Answering, Fnu Mohbat & Mohammed J. Zaki. Towards Knowledgeable Foundation Models at AAAI, 2025. [PDF](#).
9. LLaVA-Chef: A Multi-modal Generative Model for Food Recipes, Fnu Mohbat & Mohammed J. Zaki. Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM), 2024. [PDF](#).
8. Beyond Visual Augmentation: Investigating Bias in Multi-Modal Text Generation, Fnu Mohbat, Vijay Sadashivaiah, Keerthiram Murugesan, Amit Dhurandhar, Ronny Luss & Pin-Yu Chen. Fourth Workshop on Trustworthy Natural Language Processing (TrustNLP), at NAACL 2024. [PDF](#).
7. GVdoc: Graph-based Visual Document Classification, Fnu Mohbat, Mohammed J. Zaki, Catherine Finegan-Dollak & Ashish Verma. Findings of the 61st Association for Computational Linguistics (ACL), 2023. [PDF](#).
6. Teacher-Class Network: A Neural Network Compression Mechanism, Shaiq Munir Malik, Fnu Mohbat, & Murtaza Taj. The 32nd British Machine Vision Conference, 2021. [PDF](#).
5. Trash Detection on Water Channels, Mohbat Tharani, Abdul Wahab & Murtaza Taj. International Conference on Neural Information Processing, 2021. [PDF](#)
4. Cross-view Image Retrieval - Ground to Aerial Image Retrieval Through Deep Learning, Numan Khurshid, Talha Hanif, Mohbat Tharani & Murtaza Taj. International Conference on Neural Information Processing (ICONIP), Sydney, Australia. Dec 12-15, 2019. [Link](#)
3. A Residual-Dyad Encoder Discriminator Network for Remote Sensing Image Matching, Numan Khurshid*, Mohbat Tharani*, Murtaza Taj & Faisal Qureshi. IEEE Transaction on Geo-science and Remote Sensing, Nov, 2019. [Link](#)
2. Dimensionality Reduction Using Discriminative Autoencoders for Remote Sensing Image Retrieval, Mohbat, Tooba Mukhtar, Numan Khurshid & Murtaza Taj, The 20th International Conference on Image Analysis and Processing (ICAIP), Trento, Italy, September 9-13, 2019. [Link](#)
1. Use of Greendrone UAS System for Maize Crop Monitoring,

PROFESSIONAL SERVICES

- Served as a session chair at ACM Conference on Information and Knowledge Management (CIKM) 2024
- Reviewer: Computer Vision and Pattern Recognition (CVPR) 2025
- Reviewer: ACM International Conference on Information and Knowledge Management (CIKM) 2024
- Reviewer: ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD) 2024 - 2025
- Reviewer: European Conference on Computer Vision (ECCV) 2024
- Reviewer: IEEE Winter Conference on Applications of Computer Vision (WACV) 2020 - 2025

HONORS AND AWARDS

- AAAI 2025 Travel/Volunteer scholarship award (\$700 + Registration) 12/2025
- Invited talk and poster at ACM International Conference on AI in Finance (ICAIF), Brooklyn, NY. 2024
- NAACL 2024 registration award for volunteering virtual sessions 06/2024
- Founding member and Vice President, Computer Science Graduate Council at RPI 2021 - 2022
- Qualified UAV Medical Express Challenges 2018 in Australia, only 13 out of 55 teams qualified 09/2018
- Received 90% financial aid for Masters degree at Lahore University of Management Science, Pakistan 2016-2018
- National ICT R&D merit scholarship for the complete bachelor's degree. (120/7000 applicants). 2009-2014
- NESA UGRAD Exchange Scholarship (fully funded two semesters) (3/5000 applicants) 2012-2013
- Inspiring Intern certificate from the State Department during NESA exchange program (3 out of 90) 4/2013

OTHER EXPERIENCE

Rensselaer Polytechnic Institute, *Teaching Assistant*

- Computational Vision (CSCI-6270); Instructor: Chuck Stewart Spring, 2025
- Computer Science I (CSCI-1100); Instructor: Uzma Mushtaque Fall 2024
- Operating Systems (CSCI-4210); Instructor: David Goldschmidt Summer 2024
- Computer Organization (CSCI-2500); Instructor: Prof. George Slota Spring 2021

Lahore University of Management Sciences, *Teaching Assistant*

- Deep Learning (CS-437/5317); Instructor: Prof. Murtaza Taj Spring 2019, Spring 2020
- Robot Motion Planning (EE-562); Instructor: Prof. Muhammad Abubakr Fall 2018

Ivy Tech Community College *Teaching Assistant*

- Introduction to database: Oracle 11g; Instructor: Dr. John M. Heise Fall 2012
- Introduction Industrial Technology; Instructor: Dorothy R. Barse Fall 2012, Spring 2013